MD ABUL BASHAR
Network and System Administrator

RED HAT ENTERPRISE LINUX

DNS Implementation

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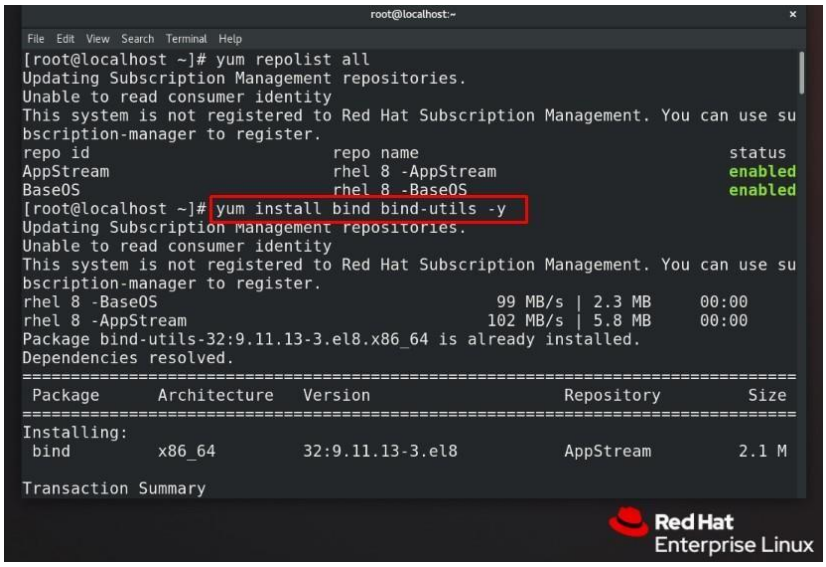
Introduction

This document explains how to set up a DNS server using BIND (Berkeley Internet Name Domain) on Red Hat Enterprise Linux (RHEL).

Implementation Guide:

Hostname: **bashar**
Server IP**: 192.168.0.10**
Domain: **bashar.com**

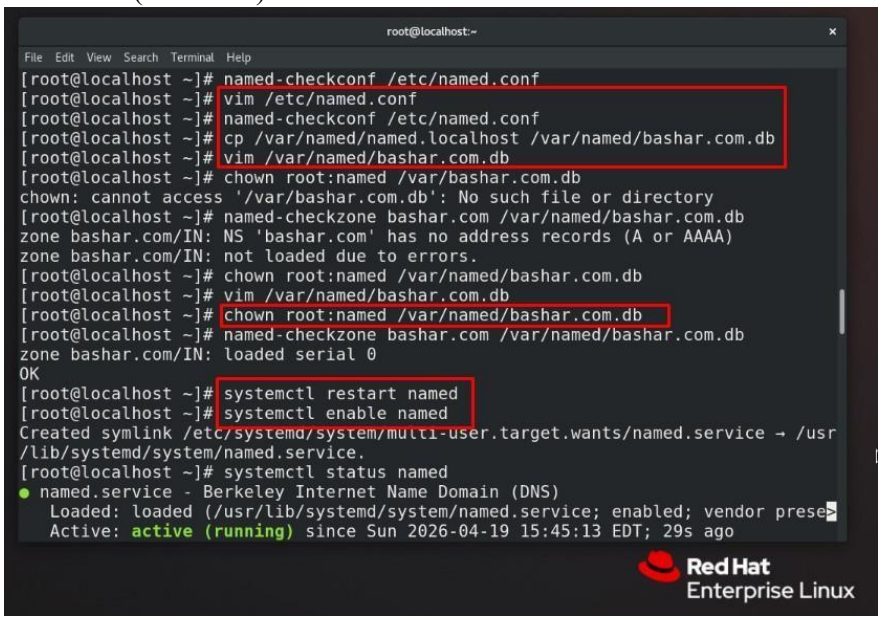
1. **Install BIND:** Run sudo dnf/yum install bind -y to get the DNS software.



2. **Edit Config:** Open **/etc/named.conf**. Add your server IP in listen-on and your network range in allow-query.

3. **Define Zone:** In the same file, add your domain name (**bashar.com**) and point it to a zone file.

4. **Create Zone File:** Go to /var/named/ and create a file for my domain. Add my server IP and Hostnames (A records) there.



```
root@localhost:~
File  Edit  View  Search  Terminal  Help
$TTL 1D
@       IN SOA  bashar.com.     root.bashar.com. (
                                        0       ; serial
                                        1D      ; refresh
                                        1H      ; retry
                                        1W      ; expire
                                        3H )    ; minimum
        IN      NS      bashar.com.
@       IN      A       192.168.0.10
bashar  IN      A       192.168.0.10
www     IN      A       192.168.0.10
~
~
~
~
~
~
~
~
~
~
~
~
                                                9,19-36       All
```

5. **Check & Start:** Use **named-checkconf** to find errors. If okay, start the service with **systemctl start named.**

```
File  Edit  View  Search  Terminal  Help
[root@localhost ~]# nslookup bashar.com
Server:         127.0.0.1
Address:        127.0.0.1#53

Name:   bashar.com
Address: 192.168.0.10

[root@localhost ~]# dig bashar.com

; <<>> DiG 9.11.13-RedHat-9.11.13-3.el8 <<>> bashar.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 38067
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
; COOKIE: f2eeb49c77f970cbfbfe71b169e5344863b1c8bcbcea8450 (good)
;; QUESTION SECTION:
;bashar.com.                    IN      A

;; ANSWER SECTION:
bashar.com.             86400   IN      A       192.168.0.10
```

6. **Firewall:** Allow DNS traffic using **firewall-cmd --add-service=dns**.

```
[root@localhost ~]# ^C
[root@localhost ~]# firewall-cmd --add-service=dns --permanent
success
[root@localhost ~]# firewall-cmd --reload
success
[root@localhost ~]#
```

2. Why RHEL DNS instead of Windows?

Many organizations choose RHEL for DNS because of these simple reasons:

- **Better Security:** RHEL is very secure. Since it doesn't need a desktop (GUI), there are fewer ways for hackers to attack.
- **Low Cost:** Windows Server requires expensive licenses for every user (CALs). RHEL (or its alternatives) is much more budget-friendly.
- **Stability:** A Linux DNS server can run for years without needing a restart. It is very lightweight and uses less RAM/CPU.
- **Independent Network:** If your Windows Server has a problem, your DNS stays alive on RHEL. This keeps your internet and apps working.
- **Easy Automation:** You can manage DNS records using simple text files, which is great for automated backups and quick changes.